

Adaptive Immersive VR Training Based on Performance and Self-Efficacy

Lasse F. Lui *

Unnikrishnan Radhakrishnan

Francesco Chinello

Konstantinos Koumaditis

Extended Reality and Robotics Lab (xR²)
Department of Business Development and Technology
Aarhus University

ABSTRACT

Effective training requires that the training experience fits the ability and confidence (i.e., self-efficacy) of the trainee. Specifically, the individual's self-efficacy should ideally be slightly higher than the difficulty of a given task. A significant benefit of immersive virtual reality (IVR) is the potential to utilize measures of trainee behavior to continuously adapt the training content to the individual. However, a major challenge involves the identification of relevant measures that can be used to adapt training content in a way that increases training output. The current paper aims to inspire further research on adaptive IVR training by describing the design and development of a study on adaptive IVR training where the use of self-efficacy measures for adaptation marks a point of departure from prior literature. The design of the proposed study is informed by analyses of results from previous IVR studies.

Index Terms: I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism—Virtual Reality;

1 INTRODUCTION

Virtual Reality (VR) refers to technologies that enable the experience of simulated reality in the form of computer-generated imagery, sounds, tactile sensations, etc. VR technologies range from desktop PC-based VR to CAVE displays and Head Mounted Displays (HMDs) [25]. The latter technologies are referred to as Immersive VR (IVR) as they immerse the user in the virtual environment by enabling wide fields of view, head tracking, and varying levels of interaction between the user and the environment. Skill training in IVR is becoming popular in the industry spanning multiple domains [20]. Interest in IVR-based skill training is primarily motivated by its potential to, among other factors, improve the learning curve, reduce training cost and time, and increase portability with safety [1]. Although previous re-vIEWS highlight most studies show positive effects of IVR training [20], the mixed and often conflicting nature of results indicates further explorations are needed to investigate when IVR training does and does not work [6]. It should be noted that non-performance related measures are also considered when measuring the effectiveness of IVR training. E.g., physiological arousal during IVR training has been linked to improvements in performance [19,21] and measures such as self-efficacy and presence (the subjective experience of being in an immersive VR environment) lead to better training outcomes, particularly when the affordances of IVR align with the used instructional method [15].

In their systematic review of IVR-based training, Abich et al. [1] identified the kinds of knowledge, skills, and abilities improved by IVR training with three distinct categories identified: psychomotor performance, knowledge acquisition, and spatial ability. Specifically, psychomotor performance in tasks such as assembly and surgery benefitted from IVR training. In their review, Radhakrishnan et

al. [20] found fine and gross motor skills to dominate publications from the medical domain. The effectiveness of IVR training is determined by varying aspects of the training environment including, among other factors, level of immersion, training method, haptic feedback, and gamification [20].

Both Abich et al. [1] and Radhakrishnan et al. [20] found time and accuracy (based on mistakes committed) as common performance metrics used to evaluate the effectiveness of VR training. Though most examples from the literature are domain-specific, especially those from the medical domain, non-domain specific examples also exist, such as assembling 3D objects, which employ a mix of procedural and fine motor skills [5, 10, 18]. The current paper extends previous non-domain specific use cases by utilizing a buzzwire fine motor skill training scenario [7, 19]. In these experiments, participants are asked to move a loop across a wire as fast as they can while making as few mistakes as possible (i.e., avoid the loop touching the wire; see Fig. 1).

Christou et al. [7] designed an IVR buzzwire task targeting movement rehabilitation among stroke-affected patients. They used this in two experiments, where the first one (among able-bodied participants) showed that hand dominance (left vs right) could be distinguished based on task performance data (such as speed and mistakes). In the second experiment, conducted among stroke patients, they were able to distinguish between the weak and the strong arms of each participant. They finally proposed an IVR buzzwire training system comprised of different levels of increasing difficulty.

Radhakrishnan et al. [19] investigated performance improvements in a physical buzzwire task among participants randomly placed into either physical training or IVR training condition (see Fig. 1). Results showed that both conditions improved task performance. Participants in the IVR condition reported more enjoyment and higher temporal demand (measured using NASA-TLX) compared to those in the physical training condition. Physiological arousal was measured using both skin conductance and heart rate sensors, and participants in the IVR condition were less aroused than those in the physical training condition. Furthermore, participants with the lowest arousal levels had higher improvements in performance than those with the highest arousal levels across conditions. Lastly, IVR training led to a statistically significant increase in self-efficacy from pre-training to post-training ($p = .016$) whereas physical training did not ($p = .545$). Thus, although IVR training did not lead to higher performance than physical training, it did lead to a significantly higher increase in self-efficacy suggesting IVR-based training is especially fruitful in supporting the trainee's self-efficacy. Based on these results, the current study explores the role and potential of self-efficacy in fine motor skills training using the buzzwire task.

Self-efficacy (SE) refers to one's perceived capabilities for learning or performing actions [22]. According to the Social Cognitive Theory formulated by Bandura [2], SE is among the most important factors in learning and training. Ideally, the individual's level of SE should be slightly higher than the difficulty of the task at hand, meaning they have the feeling of being able to complete the task without finding it too easy or too difficult for their current ability. A major source of SE is based on performance accomplishment [24], i.e. when the individual experiences success in performing an action

*e-mail: lui@btech.au.dk

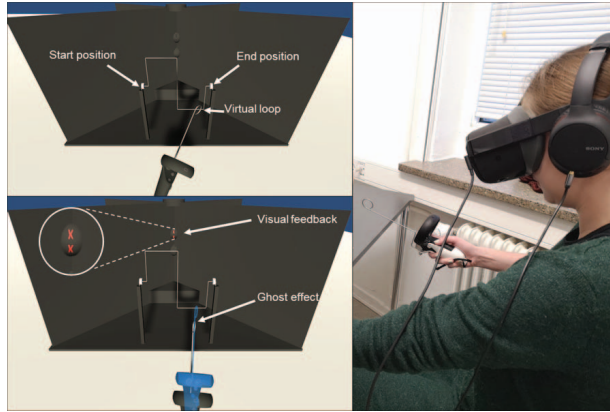


Figure 1: Right: Study participant practicing buzzwire task in VR. Top left: Virtual loop being moved across a virtual buzzwire, Bottom Left: Ghost loop appears when contact is made, and the ‘real’ loop goes outside the wire. It disappears when the loop is placed back inside the wire. Figure is taken with permission from Radhakrishnan et al. [19].

or task, their level of SE will increase. IVR training could therefore benefit from aiming to strengthen not only performance but also the individual’s confidence in performing the task at a satisfactory level (i.e., self-efficacy).

Previous research supports the importance of SE in learning and training (for more information, see the meta-analysis of 430 independent samples by Sitzmann and Ely [23]). However, few studies investigated the role of SE in IVR-based learning and training [12], with some support for the capacity of IVR to increase SE in laboratory safety training [14], crane training [24], and be as good as traditional Aviation safety training [4]. That said, studies rarely focus on how specific features of IVR can be used and changed to increase the individual’s level of SE. A central implication of viewing training through the lens of SE is that every trainee is different and has a different level of SE for the task at hand [9]. Therefore, virtual environments that adapt based on the behavior of the trainee with the aim of increasing performance through heightened SE have the potential to create a better fit between individual levels of SE and aspects of the training content.

Most current studies do not use adaptation. In other words, the same training is provided to all trainees regardless of their cognitive load, physiological arousal, and expertise level [26]. Of the few studies on adaptive IVR, most implement simple forms of adaptability, often based on pre-evaluations before or at the beginning of the IVR experience [13, 26]. Although these are partly tailored to the individual, they do not include features that continuously monitor and adapt the learning content throughout the experience. In 1969, Kelley (1969) described adaptive training as requiring that “[...] performance be continuously or repetitively measured in some way, and that the measurement be employed to make appropriate changes in the stimulus, problem, or task.” Based on a literature review, Zahabi and Abdul Razak [26] propose a framework for adaptive IVR training that incorporates performance measures and adaptive variables (i.e., changes in the training) based on adaptive logic (which changes the training based on performance measures). Inspired by Gerbaud et al. [8], Zahabi and Abdul Razak [26] describe two types of adaption based on the timing of the adaptive logic: parametering adaptation refers to adaptation occurring before the start of training, whereas dynamic adaptation refers to adaptation occurring during training. Based on such prior literature, the current paper utilizes adaptive immersive virtual reality (adaptive IVR) to refer to IVR-based applications with continuously adaptive content based on the

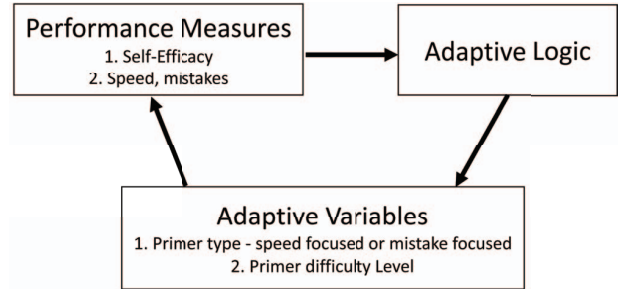


Figure 2: Design of the proposed adaptive IVR training system

real-time behavior of the user.

Adaptive IVR studies matching this definition are currently limited. For example, Kritikos et al. [11] used adaptive IVR to treat patients with arachnophobia by continuously adjusting the intensity of anxiety induction based on Electrodermal Activity (EDA) measures, whereas Muñoz et al. [17] used Electrocardiogram (ECG) and electro-encephalogram (EEG) measures to support calm-ness and cognitive readiness for military personnel training. However, the potential of adaptive IVR for fine motor skills training has not been investigated, which is the main purpose of the current paper. Nevertheless, Mariani et al. [16] provides inspiration in the form of adaptive training for robotic surgery skills in a non-immersive VR setup. In the study, participants received training in specific ‘elementary’ skills based on their performance in complex tasks during training. Despite the importance of the subjective evaluation of one’s own competence, there are currently no examples from IVR literature which forms adaptive logic based on SE. The current paper focuses on the decisions related to design, development, and testing of an adaptive IVR application that continuously adapts training content based on dynamic measures of both SE and task performance of the trainee as shown in Fig. 2. The proposed adaptive system is further detailed in the Methods section. Overall, the aim can be summed up by the following three research questions:

RQ 1: Does adaptive IVR training support performance in terms of speed and precision?

RQ 2: Does adaptive IVR training support the individual’s self-efficacy?

RQ 3: Does self-efficacy play a role in the relationship between IVR training and performance?

2 METHODS

This section describes the considerations related to designing and preparing a lab experiment based on data from previous buzzwire experiments [19, 21].

2.1 Experimental design

Approximately 120 participants are planned to be recruited for a between-subjects experiment. The study will be preregistered using AsPredicted¹ and preregistration, study materials, and data will be made available on the Open Science Framework (OSF)². As with previous buzzwire experiments, participants will complete a pre-test and post-test on a physical version of the buzzwire task, in order to measure changes in performance. Self-reported SE data will also be collected before and after training. Performance is measured in terms of task completion time (TCT), the time it takes for the loop

¹<https://aspredicted.org/>

²<https://osf.io/>

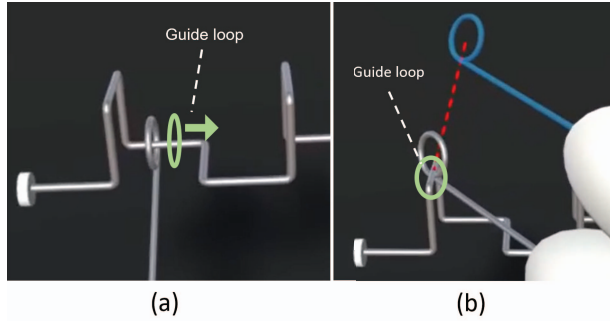


Figure 3: (a) Speed focused priming. A green guide loop moves at a constant speed across the wire. (b) Mistake-focused priming. A green guide loop stays on the wire while a mistake is being made.

to be moved from one end of the wire to the other, and *contact time* (CT), the total time the loop is in contact with the wire (as a measure of errors). During the main part of the experiment, participants will complete 10 repetitions of an IVR buzzwire task. Repetitions are important in motor skill training as according to Bernstein et al. [3]: “*Repetitions of a movement or action are necessary to solve the motor problem many times and to find the best way of solving it.*”. After each repetition, two items will be utilized to measure SE which informs the adaptive logic. Results from previous buzzwire studies (including 3 and 6 repetitions) led to the decision to use more repetitions to increase sensitivity.

It was decided to include two adaptive variables, priming type and priming difficulty, based on measures of SE and performance. In the current paper, priming refers to the guidance provided by the IVR training with the help of visual cues (elaborated in Sec. 2.2 and Sec. 2.3). To investigate the effect of adaptive training, a comparison will be made between personalized IVR training (the control group), where the training content is based on trainee behavior from the pre-test, and adaptive training (the experimental group), where training content will continuously change based on changes the trainee’s behavior.

Therefore, the type of primer and difficulty will exclusively be based on measurements from the pre-test for the control group (parameterizing adaptation), whereas performance and SE will be measured continuously to dynamically adapt the primer and difficulty throughout the training for the experimental group (dynamic adaptation). That way both groups receive a degree of adaptation, but only the experimental group will receive dynamic adaptations. The following subsections describe how the exploratory analysis was utilized to decide which measures and adaptive variables would make the best fit for the buzzwire IVR training.

2.2 Adaptive variable 1: Primer type based on self-efficacy

As described by Zahabi and Abdul Razak [26], adaptive systems should include adaptive logic based on relevant measures. Multiple regression analyses of the previous buzzwire studies indicated there may be individual differences in performance improvement. Specifically, TCT during the pre-test predicted improvements in TCT (i.e., the change from pre-test to post-test; $p < .001$, $B = 0.050$) but not CT ($p = .22$, $B = 0.018$), whereas the opposite was the case for pre-test CT, which predicted improvements in CT ($p < .001$, $B = 0.446$) but not TCT ($p = .50$, $B = 0.089$). This could indicate individual differences in the ability to perform the task either quickly or precisely predicts which type of improvement the individual trainee will achieve. Therefore, the present study will implement one of two types of priming as an adaptive variable during each IVR buzzwire

repetition to either support speed- or mistake-focused training.

For mistake-focused priming, a guide loop is added during the task. The guide loop (Fig. 3b) is always located in the optimal position of the loop, i.e., as far from the wire as possible. This is to help the participant focus on making fewer mistakes by making it apparent when they are close to making a mistake (i.e., getting further away from the guide loop).

For speed-focused priming, a guide loop (Fig. 3a) is moving through the buzzwire on its own. This is to make the participant focus on performing the task as fast as possible by competing against the guide loop.

Since indications of two different types of trainees were identified, SE was chosen as a measure for the adaptive logic since, 1) prior research supports the relevance of matching training content based on the individual’s SE, and 2) results of the study by Radhakrishnan et al. [19] suggested support for increasing SE through IVR training. Specifically, the participant will respond to two questions after each repetition of the buzzwire task (I feel confident that I can complete the task faster; I feel confident that I can complete the task with fewer mistakes). In the control group, most primers (approximately 80%) will match the type of performance the participant reports the least confidence in during the pre-test, whereas participants in the experimental group will receive the type of primer they reported the least confidence in during their last repetition of the task.

2.3 Adaptive variable 2: Primer difficulty based on performance

To support differences in baseline motor skills, priming difficulty is varied by changing the size of the loop in the mistake-focused priming and the speed of the guide loop in the speed-focused priming. Two performance measures were considered for the adaptive logic of difficulty: arousal and performance. Kritikos et al. [11] and Muñoz et al. [17] both used arousal (ECG, EDA, and EEG) in adaptive IVR. However, these studies both included stronger stimuli related to arachnophobia and military training compared to the training content in the buzzwire task. This was confirmed during exploratory analysis, where no clear pattern could be identified for using arousal to determine the adaptive logic in the buzzwire task. Instead, the performance in the pre-test was deemed more appropriate for the present design since there was a clear pattern of differences in individual performance (see multiple regression analyses described in Sec. 2.2) with a connection to the outcome measure.

In previous buzzwire studies, participant performance increased by an average of 2.25% (speed) and 3.3% (mistakes) for each repetition of the task. Based on this information, a constant difficulty will be set to match the pre-test performance for the control group followed by a constant increase in difficulty of 2% (speed) and 3% (mistakes) for each repetition. In contrast, the pre-test performance only affects the difficulty of the first repetition for the experimental group, where-after the difficulty will adapt depending on continuous measures of participant performance.

Overall, it was decided to include two measures for two types of adaptive variables to support different types of training during each buzzwire repetition: a speed- or mistake-focused primer, based on SE, and a personalized level of difficulty, based on performance.

2.4 Analysis

The main analyses of the study will be based on the three research questions:

Does adaptive IVR training support performance in terms of speed and precision?

Differences in performance will be measured as the change in performance from the pre-test to the post-test (i.e., the effect of IVR training). Previous studies utilized a relatively complex scoring

system to account for participants performing either far above or far below the mean during the pre-test. Exploratory analysis suggested that changes in performance based on percentage improvements would be a better fit for this study.

Therefore, to account for differences in baseline fine motor skill, change in performance is measured in terms of percentage improvement. For example, completion times of 60 seconds (pre) and 54 sec-onds (post) is interpreted as the same 10% improvement as completions times of 30 (pre) and 27 seconds (post). Analyses will then compare the control and experimental group in terms of mean improvements in 1) time, 2) mistakes, and 3) combined improvement.

Does adaptive IVR training support the individual's self-efficacy?

The groups will be compared in terms of mean change in SE from before to after training. Whereas two items measuring SE after each repetition of the buzzwire task are used for the adaptive logic, statis-tical analysis of the change in SE will be based on the SE measure collected before the pre- and post-test. Since the SE measure consists of multiple items, each participant's SE score is based on their mean rating of these items and change in SE will be calculated by subtracting pre-SE score from post-SE score. Secondary analysis will explore whether dynamic changes in SE during IVR training are different in the two groups.

Does self-efficacy play a role in the relationship between IVR training and performance?

The relationship between SE and training will be analyzed with multiple regression analysis predicting change in performance-based on experimental condition, SE, and their interaction. Further analysis will explore the potential mediating role of SE in the relationship between IVR-based training and improvement in performance. All main and secondary analyses will be preregistered at AsPredicted.org before data collection.

3 CONCLUSION

Adaptation presents opportunities to improve the efficacy of IVR training. While personalization using parameterized adaptive methods is predominant in the design of adaptive IVR literature, this paper presented the potential for dynamic adaptation to increase the effectiveness of training. While the current IVR adaptive training literature uses measures of task performance, cognitive load, and arousal to adapt the training, the use of self-efficacy measures for adaptive IVR training is lacking. Furthermore, there is evidence that adapting training difficulty to suit a trainee's self-efficacy leads to greater improvements in training outcomes. Three research questions were generated in order to investigate the effectiveness of an IVR fine motor skill training system that dynamically adapts training difficulty based on self-efficacy and trainee performance during training. Lastly, the planning and design of a controlled experiment were presented with the aim of answering these research questions by comparing training outcomes between IVR-based training with and without dynamically adapting content.

REFERENCES

- [1] J. Abich IV, J. Parker, J. S. Murphy, and M. Eudy. A review of the evidence for training effectiveness with virtual reality technology. *Virtual Reality*, 25(4):919–933, 2021.
- [2] A. Bandura. Social cognitive theory: An agentic perspective. *Asian journal of social psychology*, 2(1):21–41, 1999.
- [3] N. A. Bernstein, M. L. Latash, and M. T. Turvey. *Dexterity and its development*. Psychology Press, 2014.
- [4] F. Buttussi and L. Chittaro. Effects of different types of virtual reality display on presence and learning in a safety training scenario. *IEEE transactions on visualization and computer graphics*, 24(2):1063–1076, 2017.
- [5] P. Carlson, A. Peters, S. B. Gilbert, J. M. Vance, and A. Luse. Virtual training: Learning transfer of assembly tasks. *IEEE transactions on visualization and computer graphics*, 21(6):770–782, 2015.
- [6] D. Checa and A. Bustillo. A review of immersive virtual reality serious games to enhance learning and training. *Multimedia Tools and Applications*, 79(9):5501–5527, 2020.
- [7] C. G. Christou, D. Michael-Grigoriou, D. Sokratous, and M. Tsiakoulia. Buzzwirevr: An immersive game to supplement fine-motor movement therapy. In *ICAT-EGVE*, pp. 149–156, 2018.
- [8] S. Gerbaud, V. Gouranton, and B. Arnaldi. Adaptation in collaborative virtual environments for training. In *International conference on technologies for E-learning and digital entertainment*, pp. 316–327. Springer, 2009.
- [9] R. M. Klassen and E. L. Usher. Self-efficacy in educational settings: Recent research and emerging directions. *The decade ahead: Theoretical perspectives on motivation and achievement*, 2010.
- [10] K. Koumaditis, F. Chinello, P. Mitkidis, and S. Karg. Effectiveness of virtual versus physical training: The case of assembly tasks, trainer's verbal assistance, and task complexity. *IEEE Computer Graphics and Applications*, 40(5):41–56, 2020.
- [11] J. Kritikos, G. Alevizopoulos, and D. Koutsouris. Personalized virtual reality human-computer interaction for psychiatric and neurological illnesses: a dynamically adaptive virtual reality environment that changes according to real-time feedback from electrophysiological signal responses. *Frontiers in Human Neuroscience*, 15:596980, 2021.
- [12] A. Lehtikko. Measuring self-efficacy in immersive virtual learning environments: A systematic literature review. *Journal of Interactive Learning Research*, 32(2):125–146, 2021.
- [13] W. Li, H. Huang, T. Solomon, B. Esmaili, and L.-F. Yu. Synthesizing personalized construction safety training scenarios for vr training. *IEEE Transactions on Visualization and Computer Graphics*, 28(5):1993–2002, 2022.
- [14] G. Makransky, S. Borre-Gude, and R. E. Mayer. Motivational and cognitive benefits of training in immersive virtual reality based on multiple assessments. *Journal of Computer Assisted Learning*, 35(6):691–707, 2019.
- [15] G. Makransky and G. B. Petersen. The cognitive affective model of immersive learning (camil): A theoretical research-based model of learning in immersive virtual reality. *Educational Psychology Review*, 33(3):937–958, 2021.
- [16] A. Mariani, E. Pellegrini, and E. De Momi. Skill-oriented and performance-driven adaptive curricula for training in robot-assisted surgery using simulators: A feasibility study. *IEEE Transactions on Biomedical Engineering*, 68(2):685–694, 2020.
- [17] J. E. Muñoz, A. T. Pope, and L. E. Velez. Integrating biocybernetic adaptation in virtual reality training concentration and calmness in target shooting. In *Physiological Computing Systems: International Conferences, PhyCS 2016, Lisbon, Portugal, July 27–28, 2016, PhyCS 2017, Madrid, Spain, July 27–28, 2017, PhyCS 2018, Seville, Spain, September 19–21, 2018, Revised and Extended Selected Papers 3*, pp. 218–237. Springer, 2019.
- [18] M. Murcia-Lopez and A. Steed. A comparison of virtual and physical training transfer of bimanual assembly tasks. *IEEE transactions on visualization and computer graphics*, 24(4):1574–1583, 2018.
- [19] U. Radhakrishnan, F. Chinello, and K. Koumaditis. Investigating the effectiveness of immersive vr skill training and its link to physiological arousal. *Virtual Reality*, pp. 1–25, 2022.
- [20] U. Radhakrishnan, K. Koumaditis, and F. Chinello. A systematic review of immersive virtual reality for industrial skills training. *Behaviour & Information Technology*, 40(12):1310–1339, 2021.
- [21] U. Radhakrishnan, L. Kuang, K. Koumaditis, F. Chinello, and C. Pacchierotti. Haptic feedback, performance, and arousal: A comparison study in a virtual reality motor skill training task. January 2023.
- [22] D. H. Schunk and M. K. DiBenedetto. Self-efficacy theory in education. In *Handbook of motivation at school*, pp. 34–54. Routledge, 2016.
- [23] T. Sitzmann and K. Ely. A meta-analysis of self-regulated learning in work-related training and educational attainment: what we know and where we need to go. *Psychological bulletin*, 137(3):421, 2011.

- [24] H. Song, T. Kim, J. Kim, D. Ahn, and Y. Kang. Effectiveness of vr crane training with head-mounted display: Double mediation of presence and perceived usefulness. *Automation in Construction*, 122:103506, 2021.
- [25] K. Tcha-Tokey, E. Loup-Escande, O. Christmann, and S. Richir. Effects on user experience in an edutainment virtual environment: comparison between cave and hmd. In *Proceedings of the European conference on cognitive ergonomics 2017*, pp. 1–8, 2017.
- [26] M. Zahabi and A. M. Abdul Razak. Adaptive virtual reality-based training: a systematic literature review and framework. *Virtual Reality*, 24(4):725–752, 2020.